

AI Engineer Roadmap — Silki Chauhan

Generated using Chain-of-Thought Reasoning | 3rd Year B.Tech CSE Student

■ Current Position	■ Target Goal
3rd-year B.Tech CS student. Existing skills: Java, Python, SQL, Power BI, Prompt Engineering, foundational AI/ML.	Become a job-ready AI Engineer capable of building and shipping real-world AI applications within 3 months .

■ Skill Gap Analysis

Have	Need for AI Engineer role	Gap
Python, Java, SQL	Applied GenAI: LLM APIs, RAG, agents	High
Prompt Engineering basics	Production-grade prompting & eval	Medium
Power BI, data handling	Deployment (APIs, cloud, Docker basics)	High
Core AI/ML theory	Portfolio of shipped AI projects	High

■ Recommended Learning Plan	■ Suggested Projects
• Wk1-2: Python for AI, API basics (OpenAI/Anthropic SDK) • Wk3-4: Prompt engineering + evaluation techniques • Wk5-7: RAG (retrieval-augmented generation) & vector DBs • Wk8-9: Agents & tool-use / function calling • Wk10-12: Deployment basics (FastAPI, Docker, cloud hosting)	• AI-powered study assistant (RAG over notes) • Resume/portfolio analyzer using LLM API • Small AI agent automating a repetitive task • Deployed mini-app with a public demo link • Capstone: end-to-end AI app for GitHub portfolio

■ Networking Strategy

Post project progress weekly on LinkedIn (#buildinpublic), engage with AI engineers/founders, join 2-3 active AI/ML Discord or WhatsApp communities, and contribute to one open-source AI repo before month 3.

■ Monthly Milestones

Month 1	Month 2	Month 3
Python-for-AI + prompting fundamentals mastered; 1 small project shipped.	RAG + agents built; 2 projects live with public demos.	Deployment skills solid; polished portfolio + capstone project + active LinkedIn presence.

■ Immediate Next Actions

1. Get an Anthropic/OpenAI API key and run your first API call today. 2. Pick project #1 from the list above and scope it this week. 3. Create/clean up your GitHub profile as your project home base.